Code-Layout, Readability and Reusability

Date	29 June 2026
Team ID	SWTID-2026-4836
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the OptiCrop codebase in terms of structure, readability, and reusability. The project follows modular design principles using Python, Flask, and Scikit-learn, making the system maintainable and easy to extend for future agricultural prediction requirements.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used Throughout the code	Yes	Python, HTML, CSS and Flask files follow proper indentation standards.
2	Proper File Structure	Files and folders are logically organize	Yes	Project contains templates, static files, datasets, model files and backend scripts in separate folders.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables such as soil_data, crop_prediction, model, temperature and humidity are meaningfully named.
4	Function / Method Names	Functions are descriptively named	Yes	Functions like predict(), load_model(), and render_template() are clearly named.
5	Code Comments	Inline and block comments explain log	Partial	Important sections are commented, but additional comments can improve readability.
6	Modular Design	Code is split into reusable functions/m	Yes	Frontend, backend, dataset handling and prediction modules are separated.

7	No Redundant Code	Duplicate or unused code is removed	Yes	Reusable functions and libraries are used to minimize code duplication.
8	Error Handling	Exceptions and errors are handled gra	Partial	Basic input validation is implemented and can be enhanced further.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technolog	yWhere Reused	Reusability Level (High / Medium / Low)
1	Crop Prediction Module	Python, Scikit-learn	Crop recommendation prediction	High
2	Data Preprocessing Module	Python, Pandas, NumPy	Data cleaning and feature preparation	High
3	Flask Backend Module	Python, Flask	Web application routing and prediction handling	High
4	Visualization Module	Matplotlib, Seaborn	Data analysis and graphical representation	Medium
5	HTML User Interface Module	HTML, CSS	Home, About and FindYourCrop pages	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project structure and modular design.
Readability	5	Code is easy to understand and maintain.
Reusability	5	Modules can be reused in future agricultural applications.
Documentation / Comments	4	Adequate documentation with scope for additional comments.
Overall Score	5	The project follows good coding standards, modular design and maintainable architecture.